



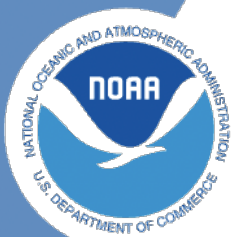
NOAA Technical Memorandum NMFS-XXX-##

# Makara Data Submission Guide

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U.S. DEPARTMENT OF COMMERCE  
National Oceanic and Atmospheric  
Administration  
National Marine Fisheries Service  
Northwest Fisheries Science Center



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# **Makara Data Submission Guide**

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# Executive Summary

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# **1 General\_Information**

## **1.1 General Information**

This guide provides data contributors with instructions, field definitions, and examples for how to enter acoustic data and metadata into the template tables and submit them for archiving in the Northeast Fisheries Science Center's (NEFSC) Makara Database. Data that are archived in Makara may also be displayed on the Passive Acoustic Cetacean Map (PACM), with permission from the data owner(s).



## 2 2\_Document\_History\_and\_Revisions

### 2.1 Document History and Revisions

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| Ver-<br>sion | Date            | Author             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|-----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0          | May 1, 2025     | Juan Lianne Wilder | Document Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.1          | June 10, 2025   | Jefrey Walker      | Updated "Data Table Template Instructions" to clarify that filenames and folder structure must match "templates" sub-directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.2          | August 26, 2025 | Jefrey Walker      | <ul style="list-style-type: none"><li>Renamed the <b>organization_code</b> to <b>deployment_organization_code</b> in the analyses and detections tables to clarify that this field refers to the organization that owns the deployment. Also added <b>analysis_organization_code</b> to the detections table. Both tables now include both the <b>deployment_organization_code</b> (organization that the deployment and recording(s) belong to), and <b>analysis_organization_code</b> (organization that the analysis belongs to).</li><li>Added <b>track_uri</b> to tracks table for linking tracks to external files in cloud storage (useful for linking track metadata to high resolution data not stored in makara)</li><li>Fixed minor typos</li></ul> |

## 2 2\_Document\_History\_and\_Revisions

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|            |                 |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.3</b> | October 7, 2025 | Jefrey Walker | <ul style="list-style-type: none"><li>• Clarified that deployments, recordings, analyses and detections are the primary data tables, but not always required for each submission</li><li>• Added <b>Sensor Datasets</b> and <b>Sensor Values</b> templates</li><li>• Added <b>deployment_realtime</b> and <b>deployment_url</b> fields</li><li>• Updated <b>JSON</b> guidelines and examples</li><li>• Set default value for <b>recording_n_channels</b> to 1</li><li>• Fixed <b>analysis_processing_types</b> to have values of POST_PROCESSED and REAL_TIME (underscores instead of dashes)</li><li>• Clarified descriptions of <b>analysis_release_data</b> and <b>analysis_release_pacm</b></li></ul> |
|------------|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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## **3 Contact\_Information\_for\_the\_Makara\_- Database\_Support\_Team**

### **3.1 Contact Information for the Makara Database Support Team**

If you have questions or issues regarding your data submission, please contact [nmfs.nec.pam-data@noaa.gov](mailto:nmfs.nec.pam-data@noaa.gov).

## 4 4\_Tips\_and\_Recommendations

### 4.1 Tips and Recommendations

- Timezones/Timestamps: All timestamps that are entered into the data submission templates should include the timezone (even if they are in UTC; e.g., 2023-10-12T18:09:05+0000). The NEFSC generally recommends that all recordings are made in UTC. Other timezones are okay, so long as the timezone information is included within the timestamp (e.g., 2023-10-12T18:09:05+0500 to indicate that the audio was recorded in Eastern Standard Time).
- Scenarios and Examples: It is highly recommended that data contributors look through our Scenarios section and the 'examples' template folder in the downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage prior to entering their data into the data submission templates. There may be an example that is similar to their situation.

## 5 5\_Data\_Table\_Template\_Instructions

### 5.1 Data Table Template Instructions

To submit data and metadata to the Makara database, the information must be organized into template tables. The data template tables and example tables are provided in a downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage, and are described in more detail below. Some of the tables are required, but do not necessarily need to be submitted at the same time. Other tables are optional, but it is highly recommended that the data contributor fills out any tables that are applicable to their deployment(s). Within each table, there are required and optional (but recommended if applicable) fields. The required tables/fields are the bare minimum of what metadata should be submitted.

Within the template zipped folder is a sub-directory named "templates/". This sub-directory contains empty CSV files that can be used as starting templates for filling out each table. Each data submission should contain one or more of these CSV files with data rows properly entered based on the formats described below. The names of each CSV file as well as the folder structure must match what is in the templates folder. Data cannot be submitted as an Excel Workbook (\*.xlsx file).

## 6 6\_Primary\_Data\_Tables

### 6.1 Primary Data Tables

The following tables are the primary tables for any data submission. Together, these four tables represent the core data stored in Makara. However, they are not strictly required for every submission. For example, a submission could contain only deployments but no recordings, analyses or detections, which could be added in a later submission. For each table, the data only needs to be submitted once unless updates or corrections are needed. In those cases, a revised table may be submitted so long as there are no changes in the primary codes identifying each row (e.g., deployment\_code, recording\_code, etc.).

| <b>Data Table</b>  | <b>Definition</b>                                                                                                     | <b>When to Submit</b>                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deployments</b> | The use of a platform (e.g., bottom-mounted mooring, buoy, glider, towed array) to record audio over a period of time | After a deployment has ended (the start date/time of a deployment is a required field, so a partial table could be submitted once a recorder is in the water, but would need to be updated after retrieval) |
| <b>Recordings</b>  | Single or multichannel recordings that were made over the course of one or more deployments                           | After a deployment has ended                                                                                                                                                                                |
| <b>Analyses</b>    | The metadata needed to understand the detection results (i.e., the parameters of the analysis protocol)               | After the data have been analyzed (if applicable, at the same time as the Detections table)                                                                                                                 |

|                   |                                                                                                                                                                                                                                               |                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Detections</b> | A time-binned or event-based identification of a sound source (e.g., dolphin, Atlantic cod, humpback whale, rain, vessel) produced from a passive acoustic detector or manual analysis that is used to determine presence of the sound source | After a recording has been analyzed (if applicable) |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|

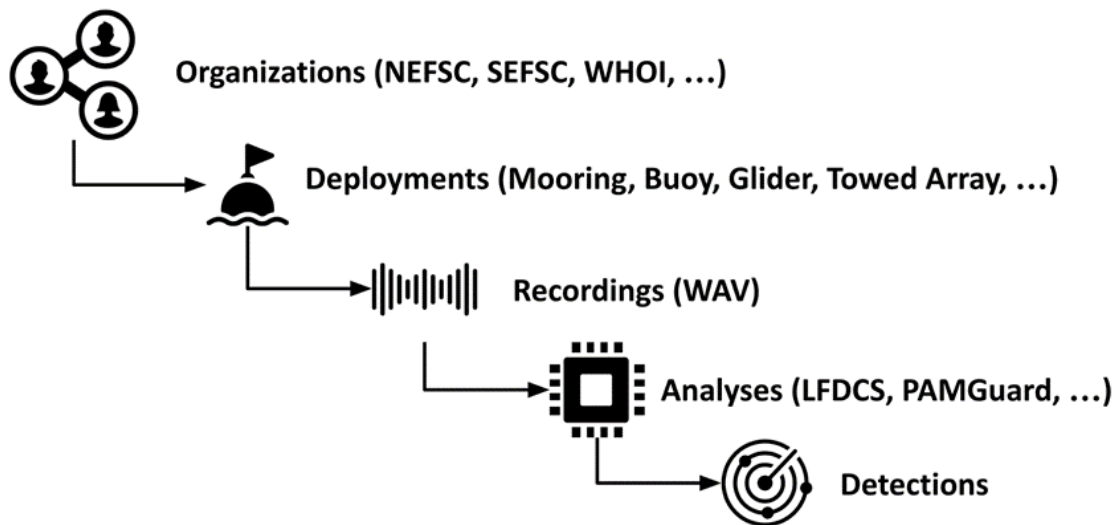


Figure 6.1: Diagram showing how the primary tables are related. Each organization owns one or more deployments, for which there can be one or more recordings. The recordings are then analyzed, generating detections.

## 7 7\_Optional\_Data\_Tables

### 7.1 Optional Data Tables (Recommended/Required if Applicable)

The following tables are optional for any data submission as long as the desired categories already exist in the database. For each table, the data needs to be submitted only once. For example, if a device has already been submitted with a deployment, and then referenced by a deployment that is submitted later, then the Device table including that device does not need to be included in the second submission as it will already exist in Makara.

| <b>Data Table</b> | <b>Definition</b>                                                                                                                                 | <b>If/When to Submit</b>                                |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Devices</b>    | A specific piece of equipment used as part of a deployment (e.g., platforms, hydrophones, recorders, sensors, acoustic releases, gliders, arrays) | If there are new devices that are not already in Makara |



## 7 7\_Optional\_Data\_Tables

|                            |                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Projects</b>            | A group of deployments                                                                                                                | If multiple deployments are submitted that the data contributor would like to have grouped together, or if new deployments are submitted that should be grouped with deployments that are already in Makara. If a project was already created with previously submitted deployment(s) and there are new deployments to be added to that project, then another Project table does not need to be submitted. |
| <b>Sites</b>               | The fixed location(s) where recordings were made over the course of one or more deployments, or general region for mobile deployments | If there are new deployment sites that are not already in Makara                                                                                                                                                                                                                                                                                                                                           |
| <b>Recording Intervals</b> | The range of time and/or frequencies with compromised or invalid audio (this table is not for tracking duty cycled data)              | If a deployment has periods of compromised data (duty cycling is accounted for in the Recordings table, not the Recording Intervals table)                                                                                                                                                                                                                                                                 |
| <b>Sensor Datasets</b>     | Metadata about the collection of sensor readings (e.g., temperatures)                                                                 | If sensor data (e.g., temperatures) were collected on the deployment                                                                                                                                                                                                                                                                                                                                       |
| <b>Sensor Values</b>       | The timestamped sensor readings belonging to a Sensor Dataset                                                                         | If sensor data (e.g., temperatures) were collected on the deployment                                                                                                                                                                                                                                                                                                                                       |

## 7 7\_Optional\_Data\_Tables

|                        |                                                                                                                            |                                     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Tracks</b>          | Metadata about the collection of track data (e.g., position, speed, and other variables) associated with a mobile platform | If a mobile deployment is submitted |
| <b>Track Positions</b> | The instantaneous timestamped location, speed, etc. along the track of a mobile platform                                   | If a mobile deployment is submitted |

## 8 8\_Reference\_Tables\_Maintained\_by\_- NEFSC

### 8.1 Reference Tables Maintained by NEFSC

NEFSC maintains the reference tables that are listed below. An updated copy of each table can be accessed by downloading the Excel document on the Passive Acoustic Reporting System Templates webpage. **If a data contributor is submitting data, but does not see the correct or applicable category that they need in one of these tables, they can contact the Makara Database Support Team to request that the category be added.** For example, if the submission includes detection data from an acoustic detector that does not already exist on the Detectors table in Makara, then the data contributor can contact the support team and have the new detector added to the reference table.

| Reference Table                   | Definition                                                                                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analysis Granularity Types</b> | The granularity of an analysis (e.g., time-binned intervals, call-by-call events, encounter events)                                          |
| <b>Analysis Processing Types</b>  | The processing method of an analysis (e.g., real-time, post-processed)                                                                       |
| <b>Analysis Quality Types</b>     | The analysis quality indicating the level of detection validation (e.g., full, partially, or not validated)                                  |
| <b>Behaviors</b>                  | The behavior associated with a detection event (e.g., foraging, socializing)                                                                 |
| <b>Call Types</b>                 | The unique call/click types associated with one or more sound sources/species (e.g., sperm whale clicks, North Atlantic right whale upcalls) |

## 8 8\_Reference\_Tables\_Maintained\_by\_NEFSC

|                                   |                                                                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Call Type / Sound Sources</b>  | A list of call type codes and sound source codes indicating which call types can be used for which sound sources. This table can be used to find the call types available for a given species by filtering for that species sound_source_code. |
| <b>Demographics</b>               | The demographics associated with a detection event (e.g., adult male, female or juvenile)                                                                                                                                                      |
| <b>Detection Quality Types</b>    | The quality of detection type indicating whether it was validated or not (e.g., unverified, valid)                                                                                                                                             |
| <b>Detection Result Types</b>     | The detection result types based on the analysis methods (e.g., detected, possibly detected, not detected)                                                                                                                                     |
| <b>Detectors</b>                  | The acoustic detector software programs used to process raw acoustic data (e.g., LFDSC, PAMGuard's whistle and moan detector)                                                                                                                  |
| <b>Device Types</b>               | The general types of devices used as part of a deployment (e.g., platforms, hydrophones, recorders, sensors, acoustic releases, gliders, arrays)                                                                                               |
| <b>Localization Methods</b>       | The methods used to localize the position of individual sound sources (e.g., DMON real-time tracking, time difference of arrival)                                                                                                              |
| <b>Localization Depth Methods</b> | The methods used to localize the depth of individual sound sources (e.g., 3D Simplex, automatic direct reflected time difference of arrival)                                                                                                   |
| <b>Organizations</b>              | Any organizations that are registered in Makara (e.g., NEFSC, Woods Hole Oceanographic Institution, Cornell University)                                                                                                                        |
| <b>Platform Types</b>             | The types of platforms upon which the hydrophones are mounted (e.g., gliders, moorings)                                                                                                                                                        |

## 8 8\_Reference\_Tables\_Maintained\_by\_NEFSC

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|                                |                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recording Quality Types</b> | The recording quality types indicate whether a recording is good, compromised, unusable, etc.                                             |
| <b>Sensor Variables</b>        | The variables that are measured by a sensor (e.g., temperature)                                                                           |
| <b>Sound Sources</b>           | Specific biological, anthropogenic, environmental, or unknown sound sources (e.g., North Atlantic right whale, recreational vessel, rain) |

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## 9 9\_Data\_Type\_Formatting\_Instructions

### 9.1 Data Type Formatting Instructions

Some fields on the data table templates require specific formatting when entering values.

| Data Type        | Required Formatting                                                                                               | Examples                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Boolean</b>   | TRUE or FALSE                                                                                                     | TRUE                                                 |
| <b>Code</b>      | Uppercase letters, numbers, underscores (_), and dashes (-) only (no spaces, lowercase letters, or other symbols) | "NEFSC_MA-RI_-202202_COX01",                         |
|                  |                                                                                                                   | "SOUNDTRAP-53423"                                    |
| <b>Code List</b> | Comma-separated list of Codes within a single string                                                              | "SOUNDTRAP-53423,VEMCO-9867,FPOD-12355"              |
| <b>Date</b>      | Date as "YYYY-MM-DD"                                                                                              | "2025-03-26"                                         |
| <b>Float</b>     | Decimal number                                                                                                    | 72.56281                                             |
| <b>Integer</b>   | Integer number                                                                                                    | 1                                                    |
| <b>JSON</b>      | Valid JavaScript Object Notation (JSON) for storing structured data objects and arrays (see JSON Instructions)    | { "KEY": "VALUE",<br>"KEY": [1, 2, 3, ...], ...<br>} |
| <b>Text</b>      | Text string (no character limit)                                                                                  | "Recorder lost at sea"                               |

---

|              |                                                                                                                                                                                                                                      |                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Times</b> | Date and time in one of the following ISO-8601 formats                                                                                                                                                                               |                                                                                     |
| <b>tamp</b>  | (seconds and milliseconds are optional), must include<br>timezone suffix (use 'Z' for UTC):                                                                                                                                          |                                                                                     |
|              | "YYYY-MM-DDTHH:MMZ"                                                                                                                                                                                                                  | "2025-03-26T15:34Z"                                                                 |
|              | "YYYY-MM-DDTHH:MM:SS±HH"                                                                                                                                                                                                             | "2025-03-26T15:34:27+04"                                                            |
|              | "YYYY-MM-DDTHH:MM:SS±HHMM"                                                                                                                                                                                                           | "2025-03-26T15:34:27-0500"                                                          |
|              | "YYYY-MM-DDTHH:MM:SS.SSS±HHMM"                                                                                                                                                                                                       | "2025-03-26T15:34:27.921-0500"                                                      |
| <b>Time-</b> | Timezone as UTC offset in one of the following formats:                                                                                                                                                                              |                                                                                     |
| <b>zone</b>  | "UTC"                                                                                                                                                                                                                                | "UTC"                                                                               |
|              | "UTC±H"                                                                                                                                                                                                                              | "UTC+4"                                                                             |
|              | "UTC±HH"                                                                                                                                                                                                                             | "UTC+04"                                                                            |
|              | "UTC±HHMM"                                                                                                                                                                                                                           | "UTC-0500"                                                                          |
| <b>URI</b>   | Valid Uniform Resource Indicator (URI) address for files<br>in cloud storage. Must start with the protocol (e.g., "gs://"<br>for Google Cloud buckets), followed by the bucket name,<br>and then path to the file within the bucket. | "gs://nefsc-1/bottom_-mounted/NEFSC_-GOM/NEFSC_GOM_-200410/NEFSC_-GOM_200410_-PU05" |
| <b>URL</b>   | Valid Uniform Resource Locator (URL) address for web<br>page                                                                                                                                                                         | "https://dcs.who.edu/gom2412/gom24we14.shtml"                                       |

---

### 9.1.1 JavaScript Object Notation (JSON) Instructions

JSON (JavaScript Object Notation) is a lightweight data interchange format that is easy for humans to read and write and for machines to parse and generate. In Makara, several fields are designated as JSON type to store structured data that doesn't fit well into standard columns. For example, information about device calibration, channel maps, and detector settings can all be stored as JSON for loading directly into other software.

### 9.1.2 When to Use JSON Fields

The JSON fields are primarily intended to be used in the following situations:

- To store structured data that benefits from consistent data structures (e.g., calibration tables)
- When you need the data to be directly queryable within the database
- When the data can be read or processed directly by other software applications

JSON is particularly valuable for device specifications, channel mappings, calibration data, and complex configuration settings. However, if your additional information is primarily free-text notes or simple comments that don't require structured querying or programmatic access, NEFSC recommends using the standard comments field available in each table instead. The comments fields are simpler to fill out, don't require special formatting, and are better suited for human-readable annotations that don't need computational processing.

### 9.1.3 What is JSON?

JSON consists of two primary structures:

- **Objects:** Collections of key-value pairs enclosed in curly braces { }. Each key must be a string in double quotes, followed by a colon :, and then a value. Example: { "DETECTOR\_THRESHOLD": 0.75 }
- **Arrays:** Ordered lists of values enclosed in square brackets [ ]. Example: [ 0.5, 0.75, 0.9 ]

Values in JSON can be:

- **Strings:** Text in double quotes "example"
- **Numbers:** Integers or decimals without quotes 42 or 3.14
- **Booleans:** true or false (lowercase, without quotes)
- **null:** Represented as null (without quotes)

There are no specific formats required for dates and timestamps. However, NEFSC recommends using the same formats listed in the Data Type Formatting Instructions table based on the ISO8601 standard (e.g., { "date": "2025-03-20", "datetime": "2025-03-26T15:34:27Z" }).



#### 9.1.4 Example JSON Structures

Simple object for the sensitivity of a recorder:

```
{ "SENSITIVITY": -1.8 }
```

Nested object for the high and low gain sensitivities of a recorder:

```
{ "GAIN": { "HIGH": 177.4, "LOW": 189.9 } }
```

Object with nested array for the sensitivity calibration of a hydrophone:

```
{ "SENSITIVITIES": [-170.1, -169.8, -170.2], "CALIBRATION_DATE": "2023-05-15" }
```

Complex nested structure containing a calibration frequency table represented as an array of objects where each object within the array is a row in the table:

```
{  
  "FREQ_TABLE": [  
    { "FREQ_HZ": 1, "GAIN_DB": 10.6 },  
    { "FREQ_HZ": 2, "GAIN_DB": 20.3 },  
    { "FREQ_HZ": 5, "GAIN_DB": 35.8 },  
    { "FREQ_HZ": 10, "GAIN_DB": 47.9 },  
    etc...  
  ]  
}
```

### 9.1.5 Recommended Guidelines

NEFSC recommends adhering to the following guidelines when creating JSON data for submission to Makara. Please note these are only guidelines, not requirements. The only requirement for these fields is that the data is in valid JSON format.

- Each entry in a JSON field should be an object at the top level, not an array or individual value so that all values have a key specifying what they are (e.g., use `{"SENSITIVITIES": [-170.1, -169.8, -170.2]}` instead of just the array of values `[-170.1, -169.8, -170.2]`)
- Always use uppercase letters for object keys to avoid changes in case across rows (e.g., `{"HIGH": 177.4}` and `{"high": 177.4}` are not compatible)
- Do not wrap numeric values in double quotes, leave them as numbers (e.g., `{"HIGH": 177.4}` instead of `{"HIGH": "177.4"}`)

### 9.1.6 Tips for Working with JSON in Spreadsheets

- Create your JSON in a text editor or JSON validator first to ensure proper formatting
- For complex JSON structures, consider using a JSON validation tool (like [jsonlint.com](https://jsonlint.com)) before pasting into your spreadsheet
- Keep JSON structures as simple as possible while still capturing the necessary data

### 9.1.7 Consistency is Critical

It is the responsibility of the data contributor to adhere to consistent structures (aka schema) within the JSON field of each table. This is critical for ensuring data integrity and usability, and so JSON data can be readily combined across multiple rows in a given table. While NEFSC provides some example formats for each JSON field in this guide, each data contributor can develop their own JSON schema that aligns with their software and workflow.

## 10 10\_Data\_Table\_Field\_Definitions

## **11 11\_Deployments\_Table\_Field\_ Definitions**

## **12 12\_Recordings\_Table\_Field\_Definitions**

## **13 13\_Analyses\_Table\_Field\_Definitions**

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